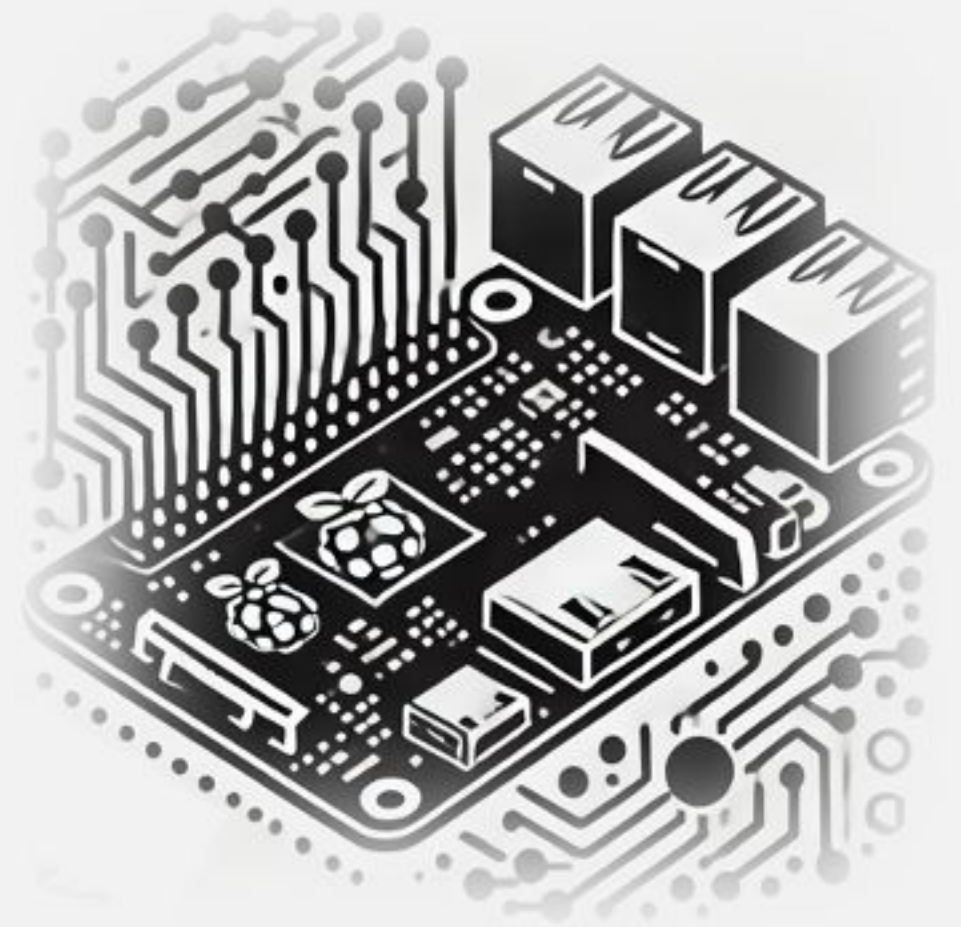


IESTI05 – Edge AI

Machine Learning System Engineering

19. SLMs: Optimization Techniques - Function Calling



Optimization Techniques:

Introduction

Techniques for Enhancing SLM at the Edge

Fundamentals: Optimizing Prompting Strategies

- Chain-of-Thought Prompting
- Few-Shot Learning
- Task Decomposition

Intermediate: Building Intelligence Systems

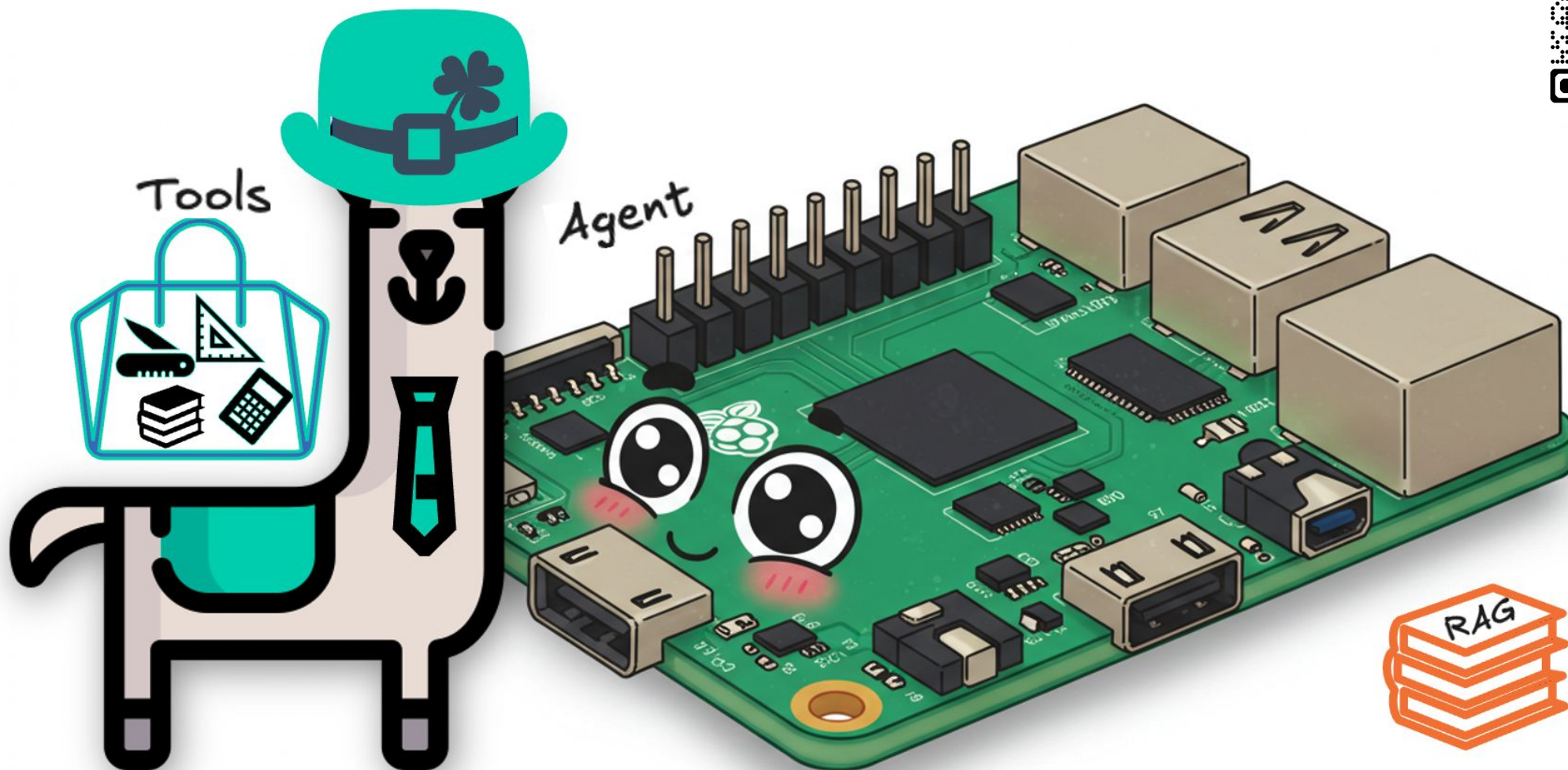
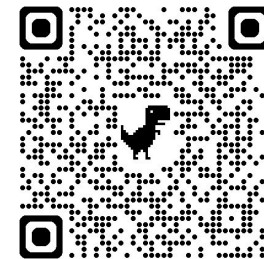
- Function Calling
- Building Agents with SLMs
- General Knowledge Router
- Response Validation

Advanced: Extending Knowledge and Specialization

- Retrieval-Augmented Generation (RAG)
- Fine-Tuning for Domain Specialization



Integration:
Combining Techniques for
Optimal Performance



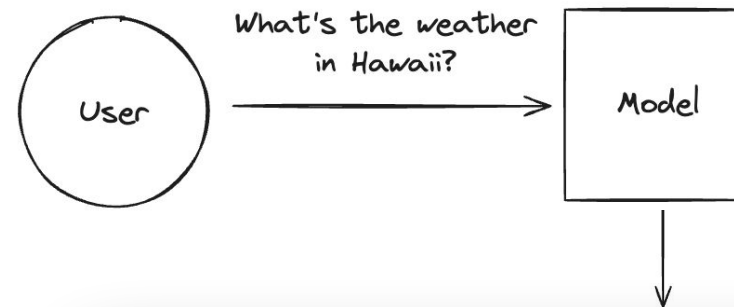
Book: Advancing EdgeAI

Function Calling

Function calling with Language Models (**LLM/SLMs**) allows these models to perform actions beyond generating text.

By integrating with external functions or APIs, LLMs can access real-time data, automate tasks, and interact with various systems.

<https://huggingface.co/docs/hugs/en/guides/function-calling>



```
python
tools = [
    {
        "type": "function",
        "function": {
            "name": "get_current_weather",
            "description": (
                "Get the current weather for a specific location. "
                "ONLY use this when users explicitly ask about weather, "
                "temperature, or climate conditions. "
                "Do NOT use for general questions about cities or locations."
            ),
            "parameters": {
                "type": "object",
                "properties": {
                    "location": {
                        "type": "string",
                        "description": (
                            "The city name, e.g., 'San Francisco', "
                            "'New York', 'London'"
                        )
                    },
                    "unit": {
                        "type": "string",
                        "enum": ["celsius", "fahrenheit"],
                        "description": (
                            "The temperature unit to use. "
                            "Defaults to celsius."
                        )
                    }
                },
                "required": ["location"]
            }
        }
    }
]
```



```
marcelo_rovai — mjrovai@raspi-5-SD: ~ — ssh mjrovai@raspi-5-SD.local — 80x21
mjrovai@raspi-5-SD:~ $ ollama run llama3.2:3b
>>> What is the distance in Km, from Santiago, Chile and the capital of France?
... Give a short answer.
The distance from Santiago, Chile to Paris, France is approximately 15,956
km (9,935 miles).

>>> What is the distance in Km, from Santiago, Chile and the capital of Colombia
... ? Give a short answer.
The distance from Santiago, Chile to Bogotá, Colombia is approximately
4,555 km (2,822 miles).

>>> What is the distance in Km, from Santiago, Chile and the capital of the USA?
... Give a short answer.
There is no single "capital" of the United States, as it has multiple
federal districts and territories, but one of the main seats of government
is Washington D.C.

The distance from Santiago, Chile to Washington D.C., United States is
approximately 7,563 km (4,696 miles).

>>> Send a message (/? for help)
```

11,647 Km

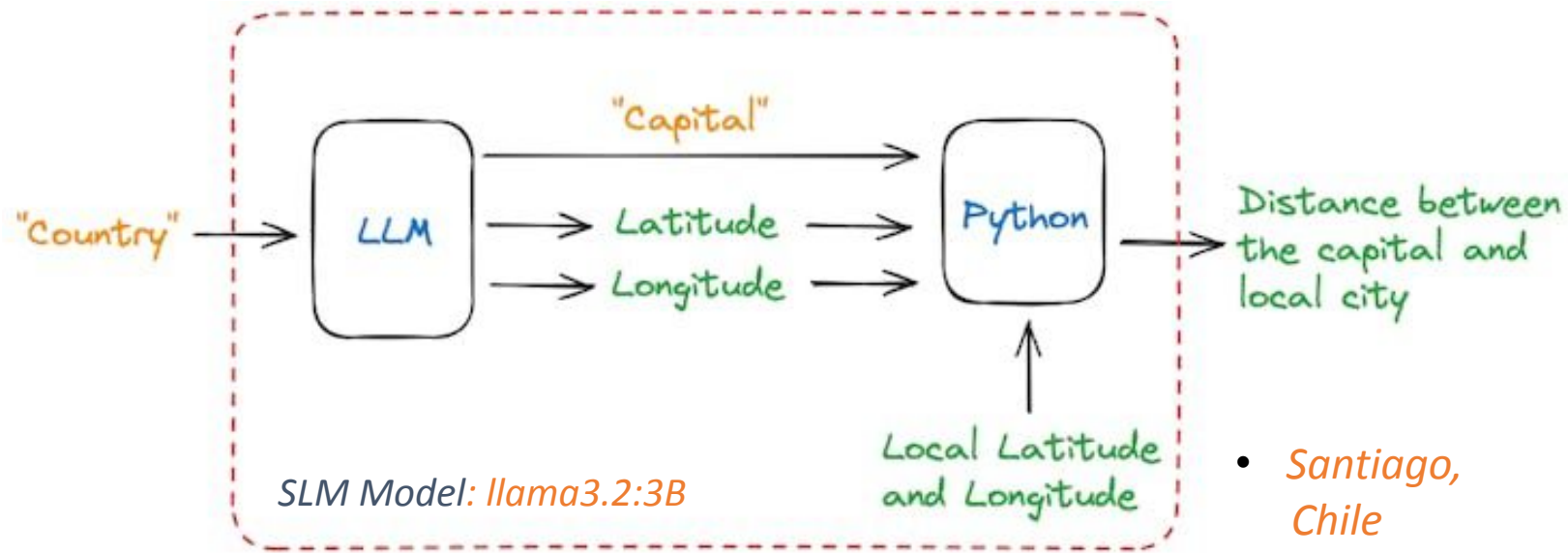
4,251 Km

8,070 Km

Wrong answers and different formats

Function Calling

- France
- Colombia
- USA



- Santiago, Chile

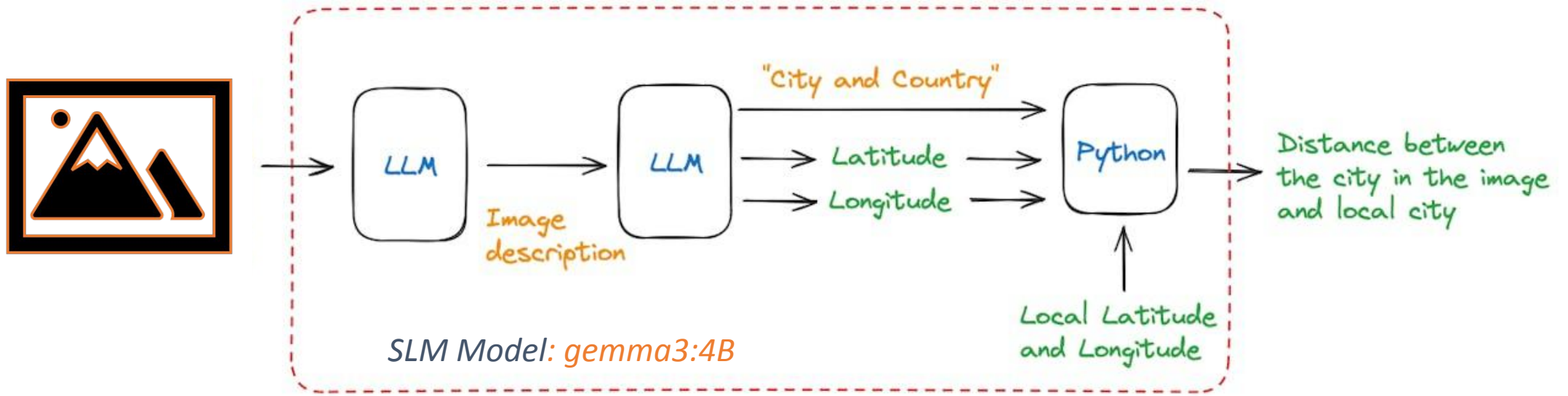
```
marcelo_rovai — mjrovai@raspi-5-SD: ~/Documents/OLLAMA/scripts — ssh mjrovai@raspi-5-SD.local — 91x11
(ollama) mjrovai@raspi-5-SD:~/Documents/OLLAMA/scripts $ python func_call_dist_calc.py

Santiago de Chile is about 11,630 kilometers away from Paris.
[INFO] ==> Model llama3.2:3B : 9.8 s

Santiago de Chile is about 4,240 kilometers away from Bogotá.
[INFO] ==> Model llama3.2:3B : 10.2 s

Santiago de Chile is about 8,240 kilometers away from Washington, D.C..
[INFO] ==> Model llama3.2:3B : 9.9 s
(ollama) mjrovai@raspi-5-SD:~/Documents/OLLAMA/scripts $
```

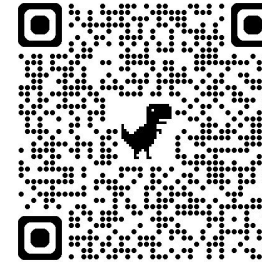
Function Calling



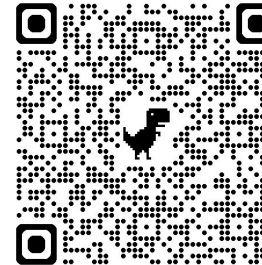
Function (Tool) Calling with Ollama



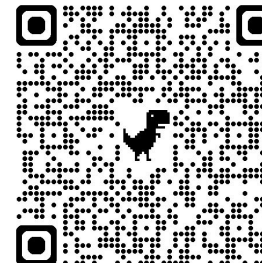
10-Ollama_Function_Calling.ipynb



20-Ollama_Function_Calling_Pydantic.ipynb



30-Function_Calling_with_images.ipynb



Questions?



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